Lecture Notes

on Probability of Compound Events, and Conditional Probability

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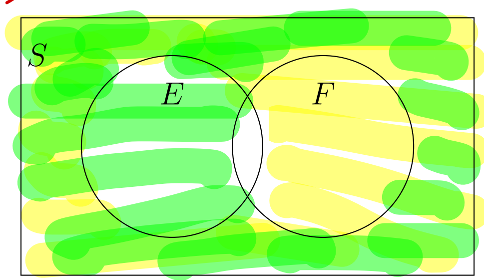
1 Properties

$P(S) = 1$
$S = E \cup \overline{E}$
$\therefore P(S) = P(E \cup \overline{E})$
$= P(E) + P(\overline{E})$
$\Rightarrow P(\overline{E}) = P(S) - P(E)$
$= 1 - P(E)$

Let E and F be two events in a sample space S.

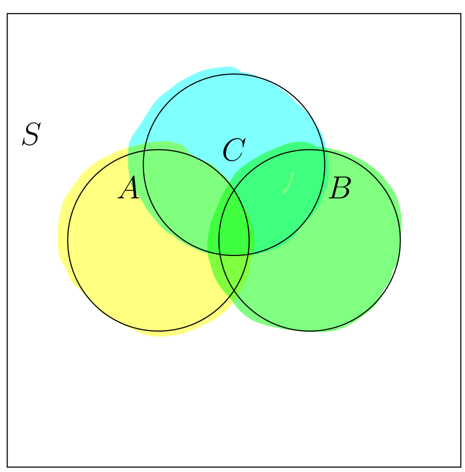
1. $P(\overline{E}) = 1 - P(E)$
2. $P(E - F) = P(E \cap \overline{F}) = P(E) - P(E \cap F)$

$P(F - E) = P(F \cap \bar{E}) = P(F) - P(E \cap F)$



3. $P(\overline{E} \cap \overline{F}) = 1 - P(E \cup F)$
4. Let $A, B,$ and C be three events in a sample space S. Then,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C).$$



2 Independent Events

Definition: Two or more events are said to be **independent** if the occurrence or non-occurrence of one event does not affect the probability of occurrence of the other.

Mathematically: For two events A and B, they are independent if

$$P(A \cap B) = P(A) \times P(B).$$

For three events A, B, and C, they are independent if

$$\begin{cases} P(A \cap B) = P(A)P(B), \\ P(A \cap C) = P(A)P(C), \\ P(B \cap C) = P(B)P(C), \\ P(A \cap B \cap C) = P(A)P(B)P(C). \end{cases}$$

Example A manufacturer of processing chips knows that 2% of its chips are defective in some way. Suppose an inspector randomly selects 4 chips for an inspection. Assuming the chips are independent, what is the probability that **at least one** of the selected chips is defective?

Let C_1, C_2, C_3, C_4 denote the 1^{st}, 2^{nd}, 3^{rd} & 4^{th} chips resp. be defective.

Then, $P(C_1) = P(C_2) = P(C_3) = P(C_4) = 0.02$.

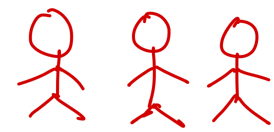
$P(\overline{C_1}) = P(\overline{C_2}) = P(\overline{C_3}) = P(\overline{C_4}) = 1 - 0.02 = 0.98$.

$$\begin{aligned} P(\text{none of the chips are defective}) &= P(\overline{C_1} \cap \overline{C_2} \cap \overline{C_3} \cap \overline{C_4}) \\ &= P(\overline{C_1})P(\overline{C_2})P(\overline{C_3})P(\overline{C_4}) \\ &= (0.98)^4 \approx 0.92 \end{aligned}$$

$$\begin{aligned} P(\text{at least one chip is defective}) &= 1 - P(\text{none of the chips are defective}) \\ &= 1 - 0.92 = 0.08. \end{aligned}$$

Example A family has 3 children. What is the probability that there is **at least one girl** among the three children?

The total number of ways of being boys and girls among 3 children $= 2 \times 2 \times 2 = 8$.



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Now, the event E, that denotes none of the children are girl.

$\therefore$ # elementary events in $E = 1$.

$$\therefore P(E) = \frac{1}{8} \quad \left[\left(\frac{\#\text{ elementary events in } E}{\#\text{ total elementary events}}\right)\right]$$

$$\therefore P(\text{at least one girl}) = 1 - P(E) = 1 - \frac{1}{8} = \frac{7}{8}.$$

Example In rolling two dice, what is the probability of rolling **a sum of** 4 **or doubles**?

Let E denotes the event of having sum of 4,
& F " " " " doubles in a rolling of two dice.

The total number of elementary events in the sample space $= 6 \times 6 = 36$.

Elementary events in the event $E = \{(1,3), (2,2), (3,1)\}$

" " " " $F = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$.

$$\therefore P(E) = \frac{3}{36} \quad \& \quad P(F) = \frac{6}{36}.$$

$E \cap F = \{(2,2)\}$

$$P(E \cap F) = \frac{1}{36}$$

$$P(E \cup F) = P(E) + P(F) - P(E \cap F) = \frac{3}{36} + \frac{6}{36} - \frac{1}{36}$$

$$= \frac{3+6-1}{36} = \frac{8}{36} = \frac{2}{9}.$$

3 Standard Deck of Cards

Summary: A standard deck of playing cards consists of **52 cards**, which are divided into **4 suits**:

$$\clubsuit \text{ (Clubs)}, \quad \spadesuit \text{ (Spades)}, \quad \heartsuit \text{ (Hearts)}, \quad \diamondsuit \text{ (Diamonds)}.$$

Each suit contains **13 cards**:

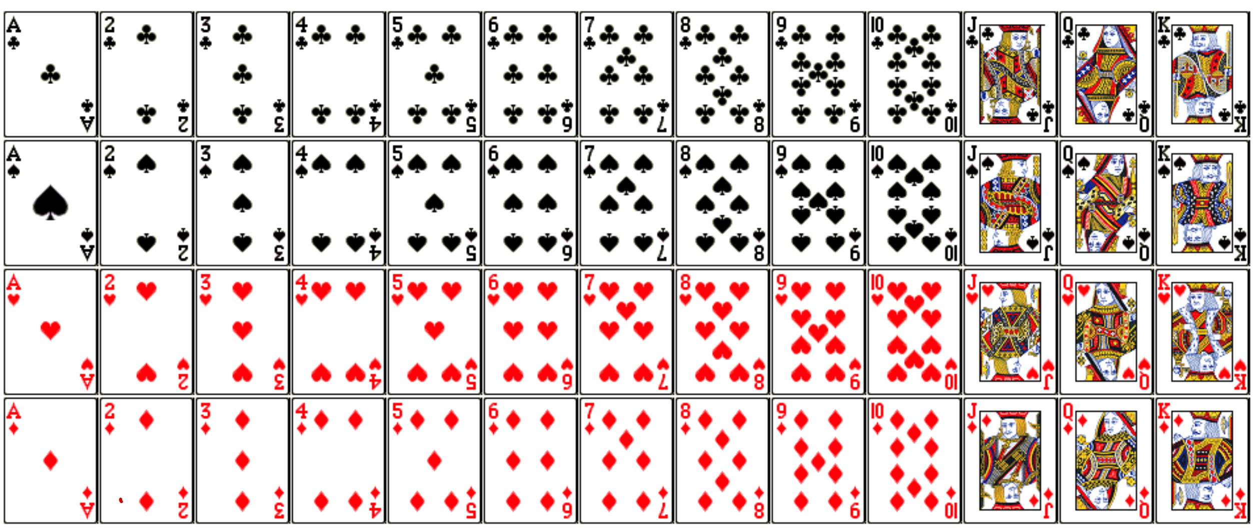
$$\text{A}, 2, 3, 4, 5, 6, 7, 8, 9, 10, \text{J}, \text{Q}, \text{K}.$$

The suits are grouped as:

$$\text{Black suits: } \clubsuit, \spadesuit \qquad \text{and} \qquad \text{Red suits: } \heartsuit, \diamondsuit.$$

Key facts used in probability:

- Total cards = 52.
- Each suit = 13 cards.
- Number of face cards (J, Q, K) = 12 (3 in each suit).
- Number of red cards = 26, number of black cards = 26.
- Probability of drawing any single specific card $= \frac{1}{52}$. $^{52}C_4$



Explanation of drawing 2 cards at a time (order do not matter) and drawing 1 by 1 (order matters).

For the first case its a combination, and the required number

$$= {}^{52}C_2 = \frac{52!}{(52-2)!\,2!} = \frac{52 \times 51 \times \cancel{50!}}{\cancel{50!}\, 2!} = \frac{52 \times 51}{2}$$

we divided by 2 as the two pairs are counted twice according to their order.

Now the case where order is important, i.e, when we choose 1 card then choose the second card, the number of ways we can do so $= {}^{52}C_1 \times {}^{51}C_1 = 52 \times 51$

(Its same to say choose 2 cards and permute) so this value will be also same to ${}^{52}P_2 = \frac{52 \times 51 \times 50!}{(52-2)!}$

$= 52 \times 51$

Example Jessica has drawn a card from a well-shuffled deck of 52 cards. Help her find the probability of the card **either being red or a king**.

R = card to be red.

K = card to be King.

Number of elementary events in the sample space $= {}^{52}C_1 = 52$.

" " " " " " event $R = {}^{26}C_1 = 26$.

" " " " " " " $K = {}^{4}C_1 = 4$.

" " " " " " " $R \cap K = {}^{2}C_1 = 2$.

$\therefore P(R) = \frac{26}{52}$, $P(K) = \frac{4}{52}$ and $P(R \cap K) = \frac{2}{52}$.

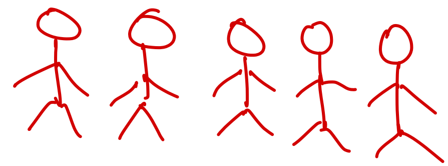
$$P(R \cup K) = P(R) + P(K) - P(R \cap K)$$

$$= \frac{26}{52} + \frac{4}{52} - \frac{2}{52}$$

$$= \frac{26 + 4 - 2}{52} = \frac{28}{52}.$$

Example A family has five children. Find the probability that there are **four girls and one boy**.

Let B and G denote the events of having Boy and having a girl.



The total number of ways to have boys and girls among 5 children

$= 2^5 = 32.$

$$\frac{5!}{(5-1)!\,1!} = \frac{5 \times \cancel{4!}}{\cancel{4!} \times 1!} = 5$$

$$P(B \cap G) = \frac{{}^5C_1 \times \underbrace{{}^4C_4}_{=1}}{2^5} = \frac{5}{32}.$$

$$\begin{cases} {}^nC_1 = n \\ {}^nC_n = 1 \end{cases}$$

$n \in \mathbb{Z}^+$